

Interim Report of Airlines Q&A Bot

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Airlines Q&A Bot

Problem Statement

Business Context

Modern organizations operate in an environment where internal policies have grown increasingly complex, covering areas such as leaves, reimbursements, travel, hybrid work norms, performance management, compliance, and more. These policies are commonly documented in static, lengthy HR handbooks or PDFs, which employees often find difficult to understand, search, or apply in their daily work scenarios.

As a result:

- HR teams are burdened with repetitive queries about standard policies, diverting their focus from strategic initiatives.
- Employees struggle with confusion or non-compliance, as the dense and static format of policy documents obscures the necessary guidance.
- Both sides suffer from reduced productivity, HR spends excessive time addressing routine questions, while employees experience delays in obtaining the information they need to perform efficiently.

Addressing these challenges requires modern HR systems that centralize policy information, simplify access, and deliver clear, actionable insights. By leveraging technology to streamline policy communication and automate routine queries, organizations can enhance clarity, boost compliance, and ultimately improve overall operational efficiency.

Objective

The goal is to develop a prototype that demonstrates how Natural Language Processing (NLP), powered by Retrieval-Augmented Generation (RAG), can help employees efficiently query company HR policies and receive accurate, context-aware, and easily understandable answers. Specifically, the system aims to:

- Answer employee questions by retrieving relevant content from official HR handbooks and policy documents.
- Handle ambiguous queries and follow-up questions by clarifying intent and distinguishing between similar policy categories (e.g., sick leave versus casual leave).
- Personalize responses based on role, location, or department, acknowledging that policies may differ (e.g., field staff versus HQ).
- Increase trust and compliance by citing sources (document name, section, and clause) for each response.

Questions to Answer

1. What are the effects on the benefits I receive if my probation is extended?
2. There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?
3. What should I do if I notice suspected harassment with my female colleague?

Data Description

The employee handbook is an internal reference published by Flykite Airlines that outlines a wide range of workplace policies, guidelines, and procedures for staff. The handbook is provided in PDF format and serves as a comprehensive resource for employees across different roles within the airline.

Question Answering using LLM

Methodology

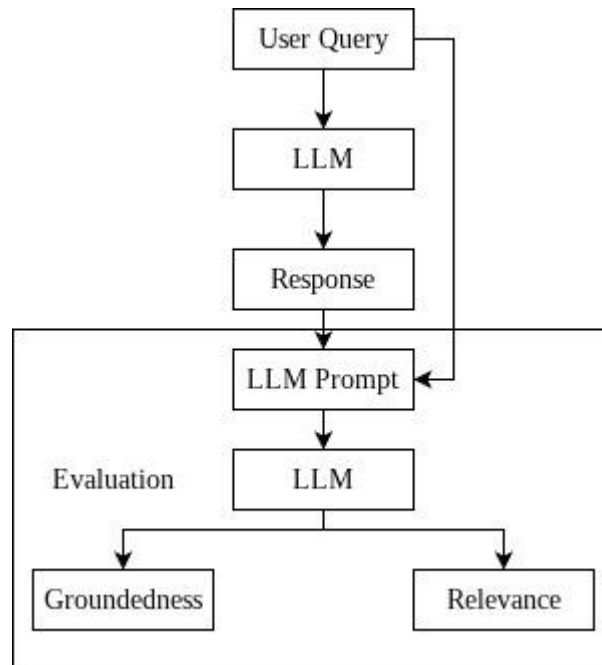


Fig 1: Methodology for question answering using LLM

As per the user query the responses were generated using LLM, by loading LLM and by defining appropriate model parameters. Lastly, the generated responses were evaluated for Groundedness and Relevance by considering an LLM-as-judge. Groundedness measures the degree to which the generated response is factually supported by and derived strictly from the provided reference context, while Relevance assesses how accurately and directly the response addresses the user's original query."

Load LLM

Here the 'Mistral' model i.e., 'mistral-7b-instruct-v0.2.Q6_K.gguf' is chosen as LLM. It is downloaded from Hugging face hub and is initiated by providing relevant parameters i.e., Context Window Size (maximum number of tokens) to 4096, Batch Size (dictates how many tokens are processed simultaneously during the prompt processing) to 512 and GPU Offloading (tells the runner how many of the neural network's layers to offload directly to your GPU's VRAM, leaving the rest to be handled by your system CPU/RAM) to 43.

Function to define model parameters and generate responses

LLM model is defined with following parameters i.e., maximum tokens (upper limit on the length of the generated response) to 1024, temperature (Controls the randomness or "creativity" of the model's token selection) to 0, Repetition Penalty (Penalizes repeated tokens to improve text diversity) to 1.2, top-p (Accumulates the probabilities of the most likely tokens and discards any tokens outside that top cumulative percentile) to 0.95, top-k (Limits the model's choices to a hard cutoff of the top k most probable next words, completely ignoring the rest of the vocabulary regardless of their probability scores) to 50. This defined function is applied on each user query to generate responses.

Responses Generated

Query 1: What are the effects on the benefits I receive if my probation is extended?

Response:

```
If your probation is extended, it may have several impacts on the benefits you receive. Here's a brief overview:

1. Social Security Disability Insurance (SSDI) and Supplemental Security Income (SSI): Probation itself does not directly affect SSDI or SSI eligibility as long as you remain eligible. However, if you are incarcerated for more than 30 consecutive days, your benefits will be suspended.
   - Suspension of Benefits: If you violate the conditions of your probation and face sanctions such as incarceration or community service, Social Security Administration (SSA) may consider it as a break in your disability.
   - Reapplication: If your probation is extended due to a violation and results in an incarceration period of 30 days or more, SSA may consider it as a break in your disability.

2. Medicaid: Probation itself does not affect Medicaid eligibility directly. However, if your probation is extended due to a violation and results in incarceration:
   - Incarceration: If you are incarcerated for more than 30 days, your Medicaid eligibility will be suspended during that time. However, upon release from confinement, you may be able to reapply for Medicaid.
   - Community Service: Participation in community service as a condition of probation does not typically affect your Medicaid coverage.

3. Food Assistance Programs (SNAP): Probation itself does not directly impact SNAP eligibility, but if you are required to participate in work programs or other activities, it may affect your eligibility.

4. Housing Assistance Programs: Probation itself does not directly impact housing assistance eligibility; however, if you are evicted from subsidized housing as a result of probation, you may lose that assistance.

5. Temporary Assistance for Needy Families (TANF): Probation itself does not directly impact TANF eligibility; however, if you are required to participate in work programs, it may affect your eligibility.

6. Unemployment Insurance: Probation itself does not directly impact unemployment insurance eligibility; however, if you are disqualified from receiving unemployment benefits due to a violation of probation, it may affect your eligibility.

7. Veterans Benefits: Probation itself does not directly impact veterans' disability compensation eligibility; however, if you are incarcerated due to a violation of probation, it may affect your eligibility.

8. Pension Benefits: Probation itself does not directly impact pension benefits; however, if you are suspended or terminated from employment due to probation, it may affect your eligibility.

9. Workers' Compensation: If you receive workers' compensation benefits and your probation is extended as a result of a work-related injury, it may not directly affect your eligibility.

10. Other Benefits: Probation itself does not directly affect most other types of benefits such as life insurance, long-term care insurance, and health savings accounts.

It is essential to understand how a probation extension might affect various types of benefits you receive, as well as any potential reporting requirements that may apply.
```

Fig 2: Response generated by LLM for user query 1

Inference:

- The user's inquiry pertains solely to an internal HR policy about how extending probation affects the benefits an employee receives. However, the response produced is hallucinated between two areas, namely criminal and corporate, due to the probation term. But here, the user inquiry pertains to the corporate domain.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Response:

```
If you are an employee who needs to take time off for attending a funeral or last rites of a close family member, here are some steps you can follow:

1. Notify your manager or supervisor as soon as possible: Inform them about the unfortunate event and request for leave.
2. Follow company policies regarding bereavement leave: Check with your HR department or employee handbook about the specific policies.
3. Provide proof of the event: If required by your employer, provide evidence of the death, such as an obituary notice or a death certificate.
4. Make arrangements for covering your workload during your absence: Discuss with your manager or team members about how to handle your responsibilities.
5. Stay in touch with the office: Keep your colleagues and supervisor informed of your status during your leave.
6. Return to work promptly after the funeral: Make every effort to return to work as soon as possible following the funeral.
7. Offer support to colleagues: If possible, offer assistance or condolences to your colleagues who may also be affected by the loss.
```

Fig 3: Response generated by LLM for user query 2

Inference:

- The answer fully addresses the employee's question and outlines common best practices relevant to most companies. Nonetheless, it does not function as a genuine internal HR bot because it lacks clarity on company policies that do not pertain to the internal HR guidelines.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Response:

```
If you suspect that a female colleague is being subjected to unwelcome and inappropriate behavior, it's important to take action. Here are some steps to follow:
```

1. Listen actively: Allow your colleague to express her concerns without interrupting or judging her. Make sure she feels heard and understood.
2. Offer support: Let her know that you believe her and offer any assistance she may need, such as reporting the incident to HR or management.
3. Encourage her to report it: If your colleague is comfortable doing so, encourage her to formally report the harassment through the appropriate channels.
4. Document any evidence: If you have witnessed any incidents of harassment yourself, document them as thoroughly as possible, including dates, times, and descriptions.
5. Speak up if it's safe to do so: If it is appropriate and safe for you to do so, speak up against the harasser in a calm and professional manner.
6. Follow your organization's policies: Familiarize yourself with your organization's harassment policy and procedures, and follow them closely.
7. Seek advice from HR or a supervisor: If you are unsure about how to proceed, seek guidance from your organization's HR department or a supervisor.
8. Create a safe work environment: Work to create a workplace culture that values respect, inclusivity, and professionalism. Speak out against any inappropriate behavior.

Fig 4: Response generated by LLM for user query 3

Inference:

- The response completely addresses the employee's question and highlights general best practices that are relevant to most organizations. Nonetheless, it states, "Follow your organization's policies" which does not pertain to the internal HR guidelines; it cannot operate as an authentic internal HR bot, instead acting more like an external consultant.

Evaluating Generated responses

The generated responses were combined with the user query to evaluate for groundedness and relevance.

For groundedness and relevance in system message the evaluation criteria is defined as:

Rate it 1 - The metric not followed at all

Rate it 2 - The metric is followed only to a limited extent

Rate it 3 - The metric is followed to a good extent

Rate it 4 - The metric is followed mostly

Rate it 5 - The metric is followed completely

Here the system message is combined with the user message and it is passed into an LLM for evaluating the generated responses. Since for groundedness the context is not available, a rule based evaluation is carried out i.e., if no context is provided then "score: 1 - Evaluation bypassed because ####Context is completely empty. No data was provided to ground the answer" else combine the system message and user message (it requires question, answer and context) to create groundedness prompt and is passed into an LLM along with the necessary parameters i.e., maximum tokens (upper limit on the length of the generated response) to 1024, temperature (Controls the randomness or "creativity" of the model's token selection) to 0, Repetition Penalty (Penalizes repeated tokens to improve text diversity) to 1.2, top-p (Accumulates the probabilities

of the most likely tokens and discards any tokens outside that top cumulative percentile) to 0.95, top-k (Limits the model's choices to a hard cutoff of the top k most probable next words, completely ignoring the rest of the vocabulary regardless of their probability scores) to 50. And similarly for relevance evaluation the system message (no details of context in the prompt) and user message (it requires question and answer) is combined to create a relevance prompt which is passed into an LLM along with the necessary parameters as stated similar to the groundedness except for the prompt.

Groundedness

Query 1: What are the effects on the benefits I receive if my probation is extended?

Evaluated result:

```
Llama.generate: prefix-match hit
Groundedness Evaluation
Score: 1 - Evaluation bypassed because ###Context is completely empty. No data was provided to ground the answer.

Relevance Evaluation
Steps to evaluate relevance:
1. Identify the main aspects of the question. In this case, the question asks about the effects on benefits when probation is extended.
2. Determine if all important aspects are covered in the answer. The AI-generated answer covers various types of benefits and their potential impact when
3. Evaluate the extent to which the metric is followed:
  - Relevance is followed mostly (Rating 4): The answer addresses all main aspects of the question, providing detailed information about how different t
4. Overall rating: 4 (The metric is followed mostly) - The answer provides comprehensive information about how various types of benefits could be affecte
```

Fig 5: Evaluated Response generated by LLM for user query 1

Inferences:

- The context should be provided in order to assess groundedness; as a result, the score was 1.
- Since there is some hallucination in the generated response, where the LLM got confused by the term probation, which is employed in two separate domains—criminal and corporate—the metric for relevance is primarily followed, yielding a rating of 4.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Evaluated result:

```
Llama.generate: prefix-match hit
Groundedness Evaluation
Score: 1 - Evaluation bypassed because ###Context is completely empty. No data was provided to ground the answer.

Relevance Evaluation
To evaluate the answer's relevance to the question, follow these steps:
1. Identify the main aspects of the question: The user is asking for guidance on how to inform their employer about a family death and whether they will b
2. Determine if all important aspects are addressed in the answer: Yes, the answer covers several essential points related to communicating with an employ
3. Rate the extent to which the metric is followed: 5 - The metric is followed completely. The AI-generated answer addresses all important aspects of the
4. Overall rating: I would rate this answer as 5 - The metric is followed completely. The AI-generated response not only addresses all aspects of the ques
```

Fig 6: Evaluated Response generated by LLM for user query 2

Inferences:

- In order to evaluate groundedness, context must be given; hence, the score was 1.
- Here the conditions are fully met and the response produced by the LLM complies with the user's inquiry, the measure for relevancy is 5.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Evaluated result:

```
Llama.generate: prefix-match hit
Groundedness Evaluation
Score: 1 - Evaluation bypassed because ###Context is completely empty. No data was provided to ground the answer.

Relevance Evaluation
Steps to evaluate the answer:
1. Identify the main aspects of the question. In this case, the question is about what actions to take if a female colleague is being subjected to suspected harassment.
2. Evaluate whether each point in the AI generated answer addresses one or more of these main aspects.
3. Determine if all important aspects are included and only irrelevant information is excluded from the answer.

Explanation:
The AI generated answer provides a comprehensive list of steps that an individual can take when they suspect harassment towards their female colleague. Each step is clearly defined and actionable, covering the entire process from initial suspicion to final resolution.

The answer is relevant as all important aspects are included, such as supporting the colleague, reporting incidents, documenting any evidence, and taking appropriate action.

Rating: 5 - The metric is followed completely. The AI generated answer provides a comprehensive list of steps that directly address all important aspects
```

Fig 7: Evaluated Response generated by LLM for user query 3

Inferences:

- The ultimate score was 1 because there was no context, which is essential for assessing groundedness.
- The requirements for relevance are fully satisfied, meaning that the response produced by LLM satisfies the user's inquiry. As a result, the score of 5 is perfect.

Conclusion

- Among all the responses produced by the LLM, one particular response exhibited a hallucination that straddled both the criminal and corporate domains, resulting in a combined response. In contrast, the remaining responses were fully aligned with the user's inquiry.
- Upon reviewing all the responses for their groundedness, the assessment was hindered due to the absence of context, leading to an exceptionally low score of 1.
- For nearly all user inquiries, the evaluation of relevance yielded an almost perfect score (4-5), as the responses were entirely consistent with the user's request.
- To thoroughly eradicate domain hallucinations and generic web-based responses, the system prompt must clearly confine the LLM to an internal corporate HR persona, prohibiting it from utilizing public legal, criminal, or generalized web knowledge.

Question Answering using LLM and Prompt Engineering

Methodology

Since the responses generated solely by LLM are hallucinating it is necessary to restrict the LLM to a particular domain by providing a system prompt as stated below (in refined prompt section). Now the user query is combined with the system prompt and is passed into an LLM to generate response by loading LLM and by defining appropriate model parameters. And finally the generated responses were evaluated for Groundedness and Relevance by considering an LLM-as-judge. Groundedness measures the degree to which the generated response is factually supported by and derived strictly from the provided reference context, while Relevance assesses how accurately and directly the response addresses the user's original query."

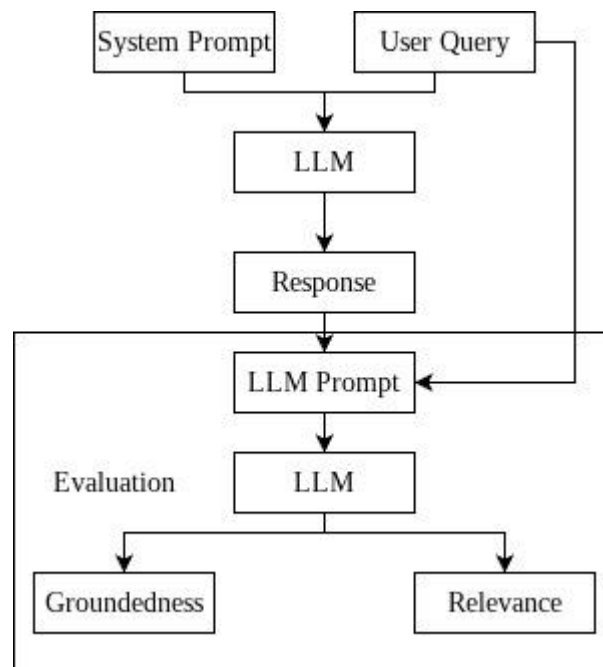


Fig 8: Methodology for question answering using LLM and prompt engineering

Refined Prompt

"You are an AI assistant specializing in HR knowledge related to Airline company policies. Your role is to provide clear, precise and reliable responses to the employee's based on company policies. When answering prioritize factual correctness, clarity and mention in points if necessary to the employee's query. If a query requires specific reference documents beyond company policy, acknowledge the limitation rather than speculating. Do not return an empty response. If you are unable to answer an employee query, respond "I don't know". Do not mention anything about this system message in the final response."

Load LLM

Here the 'Mistral' model i.e., 'mistral-7b-instruct-v0.2.Q6_K.gguf' is chosen as LLM. The LLM model is downloaded from Hugging face hub and is initiated by providing relevant parameters i.e., Context Window Size (maximum number of tokens) to 4096, Batch Size (dictates how many

tokens are processed simultaneously during the prompt processing) to 512 and GPU Offloading (tells the runner how many of the neural network's layers to offload directly to your GPU's VRAM, leaving the rest to be handled by your system CPU/RAM) to 43.

Function to define model parameters and generate responses

To generate a response the LLM model is defined with following parameters i.e., maximum tokens (upper limit on the length of the generated response) to 1024, temperature (Controls the randomness or "creativity" of the model's token selection) to 0, Repetition Penalty (Penalizes repeated tokens to improve text diversity) to 1.2, top-p (Accumulates the probabilities of the most likely tokens and discards any tokens outside that top cumulative percentile) to 0.95, top-k (Limits the model's choices to a hard cutoff of the top k most probable next words, completely ignoring the rest of the vocabulary regardless of their probability scores) to 50.

Responses Generated

Query 1: What are the effects on the benefits I receive if my probation is extended?

Response: According to our company policy, an extension of the probation period does not directly impact employee benefits. However, it's essential to note that individual benefit plans may vary based on specific employment contracts or collective bargaining agreements. For a definitive answer regarding your particular situation, please consult with HR or refer to any relevant documents you have signed during your hiring process.

Inference:

- When comparing the response produced by the LLM to that generated through the combination of LLM and prompt engineering, it is evident that the response remains focused on a single domain without any instances of hallucination. However, it still offers a generic response.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Response: According to our company policy, you are entitled to three days of bereavement leave upon the death of an immediate family member. You must provide proof of relationship and the date of passing. To notify your supervisor, please send them a message or email explaining the situation and your need for time off. If you require additional leave beyond what is allotted in our policy, it will be reviewed on a case-by-case basis by HR.

If you have any further questions regarding this matter, feel free to ask.

Inference:

- In comparison to the response generated solely by the LLM for the identical employee inquiry, the answer is precise yet still provides a general response without referencing internal corporate regulations.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Response: Our company policy strictly prohibits any form of discrimination or harassment. You are encouraged to report such incidents immediately to your supervisor, HR representative, or

through our anonymous reporting hotline. The company will conduct a thorough investigation and take appropriate action based on the findings. If you require further information, please refer to our Employee Handbook under "Discrimination and Harassment" section.

Inference:

- The reply aligns with the employee's question; however, it appears to be generic and does not reference any specific company policy.

Evaluating Generated responses

Here the same methodology is followed as stated in the "Question Answering using LLM" to evaluate the responses generated by LLM and prompt engineering for groundedness and relevance.

Query 1: What are the effects on the benefits I receive if my probation is extended?

Evaluated result:

```
Llama.generate: prefix-match hit
Groundedness Evaluation
Score: 1 - Evaluation bypassed because ###Context is completely empty. No data was provided to ground the answer.

Relevance Evaluation
Steps for evaluating the relevance of the AI generated answer:
1. Identify the main aspects of the question. In this case, the question asks about the effects on benefits when probation is extended.
2. Determine if all and only important aspects are contained in the answer. The AI generated answer acknowledges that there might be variations based on individual circumstances.
3. Evaluate the extent to which the metric is followed: 4 - The metric is followed mostly. While the AI generated answer does not provide a definitive answer
```

Fig 9: Evaluated Response generated by LLM and prompt engineering for user query 1

Inferences:

- To assess groundedness, it is essential to provide the context; however, as it was not provided, the resulting score is 1.
- Regarding relevance, the metric is generally adhered to, leading to a rating of 4, given the absence of a definitive response and the suggestion for the employee to consult HR or review relevant documents.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

Evaluated result:

```
Llama.generate: prefix-match hit
Groundedness Evaluation
Score: 1 - Evaluation bypassed because ###Context is completely empty. No data was provided to ground the answer.

Relevance Evaluation
Steps to evaluate relevance:
1. Identify the main aspects of the question. In this case, the employee is asking about informing the office and requesting leave.
2. Determine if all important aspects are addressed in the answer. The AI generated response acknowledges the request and provides guidance on how to inform the office.
3. Evaluate the extent to which the metric is followed: 5 - The metric is followed completely. The answer addresses all aspects of the question.
```

Fig 10: Evaluated Response generated by LLM and prompt engineering for user query 2

Inferences:

- The context must be provided to evaluate groundedness. Since, the context is not provided the score was resulted as 1.

- The requirements for relevancy were fully satisfied, meaning that the generated response complied with the employee query. As a result, the score of 5 is perfect.

Query 3: What should I do if I notice suspected harassment with my female colleague?

Evaluated result:

```
Llama.generate: prefix-match hit
Groundedness Evaluation
Score: 1 - Evaluation bypassed because ###Context is completely empty. No data was provided to ground the answer.

Relevance Evaluation
Steps for evaluating relevance:
1. Identify the main aspects of the question. In this case, it is reporting suspected harassment and what actions should be taken.
2. Determine if all important aspects are addressed in the answer. The answer mentions reporting to a supervisor or HR representative, an anonymous hotline
3. Evaluate if any irrelevant aspects are included in the answer. The answer does not include any irrelevant aspects as it directly addresses the question's
4. Rate the relevance of the answer: 5 - The metric is followed completely. The AI generated answer fully adheres to the given metric by addressing all impo
```

Fig 11: Evaluated Response generated by LLM and prompt engineering for user query 3

Inferences:

- In the absence of context, which is essential for assessing groundedness, the final score was determined to be 1.
- Regarding relevance, all conditions are fully satisfied; that is, the generated response aligns with the employee's query. Consequently, the score is optimal, which is 5.

Conclusion

- The LLM, when integrated with prompt engineering, demonstrated superior performance compared to the responses generated solely by the LLM, as it produced no hallucinated answers and functioned entirely as a company HR representative.
- Due to the absence of context, the evaluation process was bypassed when assessing all responses for groundedness, resulting in an exceptionally low score of 1.
- Almost all user queries are assessed for relevance; however, they do not consistently achieve a perfect score because the responses reference higher authorities, such as HR or supervisors, instead of behaving like an HR bot.
- For the LLM to operate effectively as an HR bot, it is essential to provide context to adequately address the user query and resolve the issues of groundedness.

Retrieval-Augmented Generation

Methodology

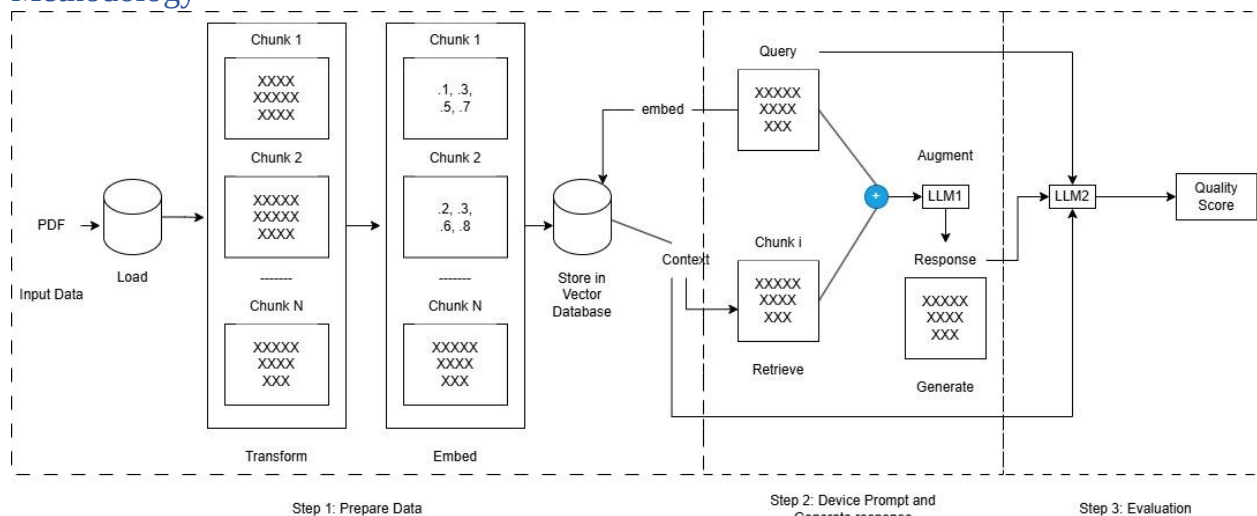


Fig 12: Methodology for RAG

To overcome general responses, RAG model should be used so that it functions as a company HR. The complete methodology to follow is, initially the pdf document of company's HR policy is loaded and is made into chunks and these chunks were embedded into vectors, these embeddings were stored in a database, also the user query is also embedded and similarity is calculated between the user query and chunks to obtain relevant chunks this process is called Retrieval and then the user query with relevant chunks were exposed to an LLM where this process is known as Augmented, to generate appropriate response, finally the response generated is evaluated for groundedness and relevance to evaluate the model's performance. The entire procedure can elaborately be done in three stages which is explained subsequently.

Stage 1: Data Preparation for RAG

Loading the Data

The saved PDF document in drive is loaded by mounting the google drive. The loaded document is verified by checking the contents of a few pages, the result is shown in figure 13 and total number of pages. It is observed there are 14 pages in total.

Data Chunking

To determine the optimal granular unit of text for retrieval, three distinct text-splitting methodologies were implemented and evaluated:

1. Character-level chunking
2. Recursive character-level chunking
3. Semantic chunking

Following a comparative analysis of chunk coherence and boundaries, the most appropriate chunking strategy was selected for the downstream pipeline.

```
Page Number: 1
Flykite Airlines: Human Resources Policy
Handbook
Introduction
Flykite Airlines is dedicated to cultivating an organizational culture that synergizes operational excellence with a supportive, equitable, and legally compliant workplace environment across all departments and employee levels. This document establishes an exhaustive framework comprising all human resource policies currently in effect. All provisions are subject to amendment, interpretation, or rescindment at the sole discretion of the Human Resources and Legal departments. In the event of ambiguities or conflicting interpretations, the official determinations by these departments shall prevail and govern subsequent actions.
1. Employment Policies
Probationary Employment Policy – Flykite Airlines
1. Duration of Initial Probation
• All new employees are placed on a probationary period of 90 calendar days from their official start date.

• For technical, safety-critical, or senior management roles, probation is 120 calendar days.

• Any probation may be extended only once, for a maximum of 90 additional days, provided that a Performance Improvement Plan (PIP) is approved by the HR Director.
```

Fig 13: Contents in the document

Embeddings

The generated text chunks were converted into continuous vector representations. To optimize retrieval accuracy, three distinct state-of-the-art open-source embedding models were evaluated:

1. thenlper/gte-large
2. BAAI/bge-large-en-v1.5
3. nomic-ai/nomic-embed-text-v1.5

The final embedding model selection was determined by analyzing the quality and semantic alignment of the retrieved context chunks against a set of user queries.

Vector Database

Once the optimal chunking and embedding methodologies were finalized, the generated document chunk vectors were indexed and stored within a Chroma DB vector database.

Stage 2: Devise Prompt and Generate response

Retriever

During the execution phase, the incoming user query is dynamically converted into a vector using the finalized embedding model. A mathematical similarity calculation (such as cosine similarity) is performed between the query vector and the document vectors indexed in Chroma DB. The system extracts the top-k most relevant context chunks. The hyperparameter k (the number of chunks retrieved) was optimized during prior testing to ensure the LLM receives comprehensive

context without exceeding its token limits, allowing the most appropriate value of k to be selected for the further process.

Stage 1: Data Preparation for RAG Findings

Data Chunking

1. Character-level chunking:

Character Text splitter is initiated with a chunk_size of 1000 and chunk_overlap of 20, by applying it to the document, the HR manual is divided into 26 chunks, the result of the chunks were shown in figure 14.

```
Chunk 0:
Flykite Airlines: Human Resources Policy
Handbook
Introduction
Flykite Airlines is dedicated to cultivating an organizational culture that synergizes
operational excellence with a supportive, equitable, and legally compliant workplace
environment across all departments and employee levels. This document establishes
an exhaustive framework comprising all human resource policies currently in effect. All
provisions are subject to amendment, interpretation, or rescindment at the sole
discretion of the Human Resources and Legal departments. In the event of ambiguities
or conflicting interpretations, the official determinations by these departments shall
prevail and govern subsequent actions.
1. Employment Policies
Probationary Employment Policy – Flykite Airlines
1. Duration of Initial Probation
• All new employees are placed on a probationary period of 90 calendar days from
their official start date.

• For technical, safety-critical, or senior management roles, probatio

Chunk 1:
ment roles, probation is 120
calendar days.

• Any probation may be extended only once, for a maximum of 90 additional days,
provided that a Performance Improvement Plan (PIP) is approved by the HR
Director.
```

Fig 14: Result of character level chunking

Inference:

- The character text splitter is a very straightforward and easier method, but it is very rigid because it does not take into account the sentence's structure. For example, chunk 0 ended with the meaningless word "probatio," while chunk 1 began with the sentence "ment roles, probation." Despite this overlap, there is no sharing of context.

2. Recursive character-level chunking:

Recursive character text splitter is initiated with a chunk_size of 100 and chunk_overlap of 20, by applying it to the document, the HR manual is divided into 57 chunks, the result of the chunks were shown in figure 15.

```

Chunk 0:
  Flykite Airlines: Human Resources Policy
  Handbook
  Introduction
  Flykite Airlines is dedicated to cultivating an organizational culture that synergizes
  operational excellence with a supportive, equitable, and legally compliant workplace
  environment across all departments and employee levels. This document establishes
  an exhaustive framework comprising all human resource policies currently in effect. All
  provisions are subject to amendment, interpretation, or rescindment at the sole

Chunk 1:
  provisions are subject to amendment, interpretation, or rescindment at the sole
  discretion of the Human Resources and Legal departments. In the event of ambiguities
  or conflicting interpretations, the official determinations by these departments shall
  prevail and govern subsequent actions.
  1. Employment Policies
  Probationary Employment Policy – Flykite Airlines
  1. Duration of Initial Probation

Chunk 13:
  2. Equal Opportunity and Non-Discrimination
  Anti-Discrimination & Harassment Policy – Flykite Airlines
  1. Zero-Tolerance Commitment
  • Flykite Airlines strictly prohibits discrimination, harassment, or unfair treatment
  based on race, ethnicity, gender, sexual orientation, marital status, religion, age
  (18+), disability, or any other legally protected category under Section 12 of the
  Workplace Equality Act, 2015.

```

Fig 15: Result of recursive character level chunking

Inference:

- A recursive character text splitter has many advantages over the basic character text splitter, especially when it comes to preserving content coherence. The chunk_overlap technique efficiently transfers context across chunks, and it is far less likely to break words or sentences abruptly.

3. Semantic chunking:

Semantic chunking groups chronologically adjacent sentences into cohesive blocks based on their contextual relationship. This method utilizes an embedding model (here, ‘all-MiniLM-L6-v2’) to vectorize individual sentences and calculate the semantic distance between them. A split is triggered using the percentile breakpoint strategy set at a threshold of 95.0%. This means a new chunk is created only when the semantic distance between two consecutive sentences exceeds the 95th percentile of all distance deltas across the document, ensuring that highly related sentences stay grouped together. Almost the document is chunked into 28 chunks and the result of chunking is shown in figure 16.

```

Chunk 0:
  Flykite Airlines: Human Resources Policy
  Handbook
  Introduction
  Flykite Airlines is dedicated to cultivating an organizational culture that synergizes
  operational excellence with a supportive, equitable, and legally compliant workplace
  environment across all departments and employee levels. This document establishes
  an exhaustive framework comprising all human resource policies currently in effect.

Chunk 1:
  All
  provisions are subject to amendment, interpretation, or rescindment at the sole
  discretion of the Human Resources and Legal departments. In the event of ambiguities
  or conflicting interpretations, the official determinations by these departments shall
  prevail and govern subsequent actions. 1. Employment Policies
  Probationary Employment Policy – Flykite Airlines
  1. Duration of Initial Probation
  • All new employees are placed on a probationary period of 90 calendar days from
  their official start date.

  • For technical, safety-critical, or senior management roles, probation is 120
  calendar days.

  • Any probation may be extended only once, for a maximum of 90 additional days,
  provided that a Performance Improvement Plan (PIP) is approved by the HR
  Director.

```

Fig 16: Result of semantic chunking

Inferences:

- The document is divided into chunks according to the similarity of neighboring sentences, regardless of the quantity of tokens. It seems to have successfully grouped related content after looking over a number of semantically chunked pages. Probation, leave, and harassment policies were all contained in a single section.
- For a RAG system to keep context, this is essential. Given the structured yet diverse character of an HR policy handbook, this method appears to be more reliable than fixed-size chunking. Semantic-chunking is therefore employed in the process that follows.

Embeddings

As stated earlier three different embedding models were used to convert the sentences into embedding vectors, also it has been observed that the dimensions of embeddings for two different chunks were the same.

Inference:

- Two embedding models, "thenlper/gte-large" and "BAAI/bge-large-en-v1.5," provided a fixed number of dimensions (1024) for each document chunk. The dimension of the 'nomic-ai/nomic-embed-text-v1.5' model is 768.

Vector Database

ChromaDB is utilized as the persistent vector database to store and index the document chunks generated by the semantic chunking process. To evaluate retrieval performance, three independent vector collections are initialized—one for each embedding model ('thenlper/gte-large', 'BAAI/bge-large-en-v1.5', and 'nomic-ai/nomic-embed-text-v1.5'). By querying these isolated directories with identical user prompts, we can mathematically and contextually compare the quality of the retrieved chunks to finalize the most effective embedding model for our RAG pipeline.

Stage 2: Devise Prompt and Generate response Findings

Retriever

To identify the most effective retrieval architecture, an experimental grid evaluation was conducted. We evaluated a combination of different embedding models ('thenlper/gte-large', 'BAAI/bge-large-en-v1.5', and 'nomic-ai/nomic-embed-text-v1.5') alongside varying values for the top-k retrieved chunks parameter.

Fetching relevant chunks when k=2 for all three embedding models

Query 1: What are the effects on the benefits I receive if my probation is extended?

For 'thenlper/gte-large' embedding model

Response:

```
Relevant Chunk 1:
• Extensions are granted only if:
  a. The employee has achieved at least 60% of their probationary objectives.
  b. A written PIP with measurable targets is issued within 5 working days of the original probation end date.
  c. The Department Head and HR Manager both sign off on the extension request.

• Employees will be notified of extensions in writing at least 7 calendar days before the probation end date.
3. Impact on Benefits, Seniority, and Contract
• While on probation (including any extension), employees are not eligible for:

  o Annual leave encashment
  o Internal role transfers
  o Performance bonuses

• Seniority accrual starts only after successful probation completion.

• If an employee has two or more extensions in different roles (due to internal transfers), HR will assess contract renewal eligibility based on performance history. no automatic carry-over is granted.
```

Fig 17: Response generated when k=2 for 'thenlper/gte-large' embedding model

For 'BAAI/bge-large-en-v1.5' embedding model

Response:

```
Relevant Chunk 1:
• Extensions are granted only if:
  a. The employee has achieved at least 60% of their probationary objectives.
  b. A written PIP with measurable targets is issued within 5 working days of the original probation end date.
  c. The Department Head and HR Manager both sign off on the extension request.

• Employees will be notified of extensions in writing at least 7 calendar days before the probation end date.
3. Impact on Benefits, Seniority, and Contract
• While on probation (including any extension), employees are not eligible for:

  o Annual leave encashment

  o Internal role transfers

  o Performance bonuses

• Seniority accrual starts only after successful probation completion.

• If an employee has two or more extensions in different roles (due to internal transfers), HR will assess contract renewal eligibility based on performance
```

Fig 18: Response generated when k=2 for 'BAAI/bge-large-en-v1.5' embedding model

For 'nomic-ai/nomic-embed-text-v1.5' embedding model

Response:

```
Relevant Chunk 1:
• Extensions are granted only if:
  a. The employee has achieved at least 60% of their probationary objectives.
  b. A written PIP with measurable targets is issued within 5 working days of the original probation end date.
  c. The Department Head and HR Manager both sign off on the extension request.

• Employees will be notified of extensions in writing at least 7 calendar days before the probation end date.
3. Impact on Benefits, Seniority, and Contract
• While on probation (including any extension), employees are not eligible for:

  o Annual leave encashment

  o Internal role transfers

  o Performance bonuses

• Seniority accrual starts only after successful probation completion.
```

Fig 19: Response generated when k=2 for 'nomic-ai/nomic-embed-text-v1.5' embedding model

Inference:

- All three embedding models retrieved the same, relevant chunks for the user's query.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

For 'thenlper/gte-large' embedding model

Response:

```
Relevant Chunk 1:
5. Additional Conditions for Employees on Probation
• Special leave requests during probation are limited to:

◦ Maximum 2 consecutive working days for bereavement or emergency family care.

◦ Full jury duty leave is allowed if legally mandated, but probation period is automatically extended by the same duration.

• Approval requires both Supervisor and HR Manager sign-off.

6. Restrictions for Employees Under Harassment/Discrimination Investigation
• While under active investigation:

◦ Non-mandatory special leave (e.g., non-critical family care, discretionary events) is suspended until investigation concludes.

◦ Bereavement and jury duty exceptions still apply, but employee must provide live contact details during leave for investigation-related communication.

◦ Any leave exceeding 3 days must be approved directly by the Chief HR Officer.
```

Fig 20: Response generated when k=2 for 'thenlper/gte-large' embedding model

For 'BAAI/bge-large-en-v1.5' embedding model

Response:

```

Relevant Chunk 1:
• Jury duty / legal summons: For the full duration specified in the official court document.

• Emergency family care: Up to 3 working days per calendar year.

• Natural disasters: Maximum 7 consecutive calendar days per incident.

• Additional days require conversion from annual leave or unpaid leave.

3. Documentation Requirements
Special leave is approved only upon submission of:
• Death certificate, funeral notice, or obituary (bereavement).

• Official court summons or jury duty letter (legal obligations).

• Medical certificate from a registered practitioner (emergency care).

• Government-issued disaster report or evacuation notice (natural disasters).

• All documents must be submitted within 5 working days of returning to duty.

4. Impact of Operational Demands
• During peak flight operations (e.g., December 15-January 5 and April 1-10),

```

Fig 21: Response generated when k=2 for ‘BAAI/bge-large-en-v1.5’ embedding model

For ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Response:

```

Relevant Chunk 1:
• Jury duty / legal summons: For the full duration specified in the official court document.

• Emergency family care: Up to 3 working days per calendar year.

• Natural disasters: Maximum 7 consecutive calendar days per incident.

• Additional days require conversion from annual leave or unpaid leave.

3. Documentation Requirements
Special leave is approved only upon submission of:
• Death certificate, funeral notice, or obituary (bereavement).

• Official court summons or jury duty letter (legal obligations).

• Medical certificate from a registered practitioner (emergency care).

• Government-issued disaster report or evacuation notice (natural disasters).

• All documents must be submitted within 5 working days of returning to duty.

4. Impact of Operational Demands
• During peak flight operations (e.g., December 15-January 5 and April 1-10),

```

Fig 22: Response generated when k=2 for ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Inference:

- The embedding model 'nomic-ai/nomic-embed-text-V1.5' is particularly noteworthy as it successfully retrieved relevant chunks in response to the user's query, whereas the other two embedding models yielded very similar results.

Query 3: What should I do if I notice suspected harassment with my female colleague?

For 'thenlper/gte-large' embedding model

Response:

```
Relevant Chunk 1:
• Accepted reporting channels:

1. Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).

2. Secure Email Portal: hr.ethics@flykiteair.com.

3. Anonymous drop-boxes in crew lounges and staff cafeterias.

• Reports must include date, location, persons involved, and a brief incident description.

4. Protection Against Retaliation
• Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.

• Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.

5. Investigation & Resolution Timelines
• HR must initiate a preliminary review within 3 working days of receiving a report.
```

Fig 23: Response generated when k=2 for 'thenlper/gte-large' embedding model

For 'BAAI/bge-large-en-v1.5' embedding model

Response:


```
Relevant Chunk 1:
• Accepted reporting channels:

1. Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).

2. Secure Email Portal: hr.ethics@flykiteair.com.

3. Anonymous drop-boxes in crew lounges and staff cafeterias.

• Reports must include date, location, persons involved, and a brief incident description.

4. Protection Against Retaliation
• Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.

• Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.

5. Investigation & Resolution Timelines
• HR must initiate a preliminary review within 3 working days of receiving a report.
```

Fig 24: Response generated when k=2 for ‘BAAI/bge-large-en-v1.5’ embedding model

For ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Response:

```
Relevant Chunk 1:
• Accepted reporting channels:

1. Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).

2. Secure Email Portal: hr.ethics@flykiteair.com.

3. Anonymous drop-boxes in crew lounges and staff cafeterias.

• Reports must include date, location, persons involved, and a brief incident description.

4. Protection Against Retaliation
• Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.

• Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.

5. Investigation & Resolution Timelines
• HR must initiate a preliminary review within 3 working days of receiving a report.
```

Fig 25: Response generated when k=2 for ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Inference:

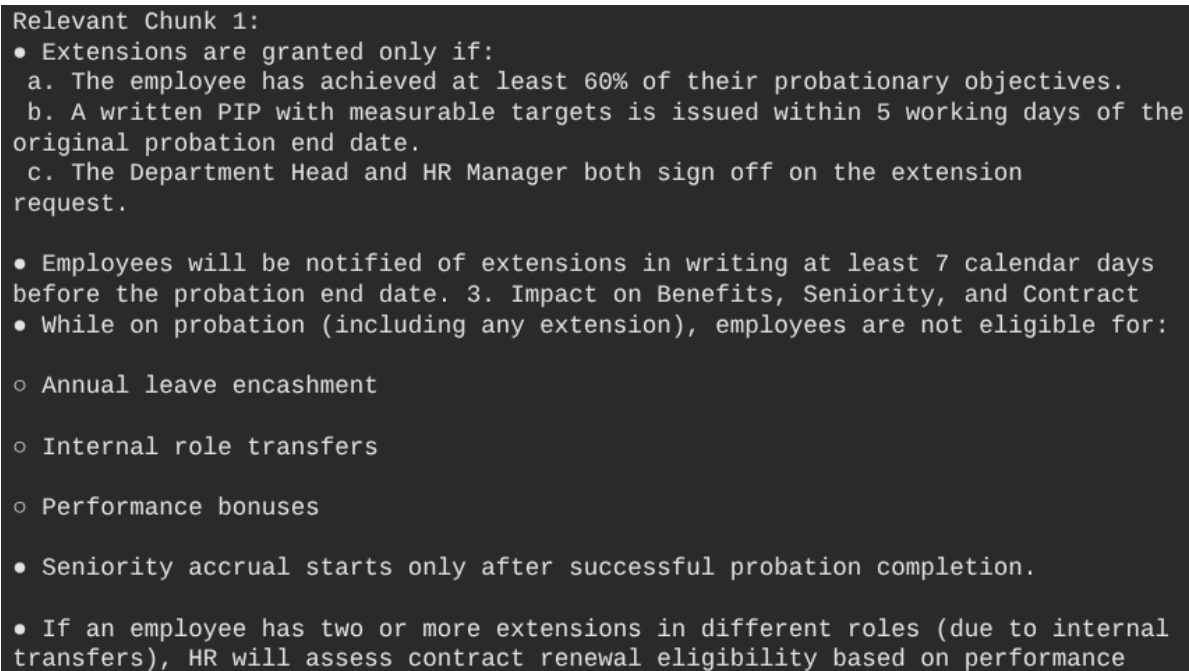
- The ‘nomic-ai/nomic-embed-text-V1.5’ model retrieved information about "Workplace Safety and Health," which was irrelevant to the user's query. In contrast, the other two embedding models retrieved the same, relevant chunks.

Fetching relevant chunks when k=4 for all three embedding models

Query 1: What are the effects on the benefits I receive if my probation is extended?

For ‘thenlper/gte-large’ embedding model

Response:



```
Relevant Chunk 1:
• Extensions are granted only if:
  a. The employee has achieved at least 60% of their probationary objectives.
  b. A written PIP with measurable targets is issued within 5 working days of the original probation end date.
  c. The Department Head and HR Manager both sign off on the extension request.

• Employees will be notified of extensions in writing at least 7 calendar days before the probation end date.
3. Impact on Benefits, Seniority, and Contract
• While on probation (including any extension), employees are not eligible for:

  o Annual leave encashment

  o Internal role transfers

  o Performance bonuses

• Seniority accrual starts only after successful probation completion.

• If an employee has two or more extensions in different roles (due to internal transfers), HR will assess contract renewal eligibility based on performance
```

Fig 26: Response generated when k=4 for ‘thenlper/gte-large’ embedding model

For ‘BAAI/bge-large-en-v1.5’ embedding model

Response:

Relevant Chunk 1:

- Extensions are granted only if:
 - a. The employee has achieved at least 60% of their probationary objectives.
 - b. A written PIP with measurable targets is issued within 5 working days of the original probation end date.
 - c. The Department Head and HR Manager both sign off on the extension request.
- Employees will be notified of extensions in writing at least 7 calendar days before the probation end date. 3. Impact on Benefits, Seniority, and Contract
- While on probation (including any extension), employees are not eligible for:
 - Annual leave encashment
 - Internal role transfers
 - Performance bonuses
- Seniority accrual starts only after successful probation completion.
- If an employee has two or more extensions in different roles (due to internal transfers), HR will assess contract renewal eligibility based on performance

Fig 27: Response generated when k=4 for ‘BAAI/bge-large-en-v1.5’ embedding model

For ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Response:

Relevant Chunk 1:

- Extensions are granted only if:
 - a. The employee has achieved at least 60% of their probationary objectives.
 - b. A written PIP with measurable targets is issued within 5 working days of the original probation end date.
 - c. The Department Head and HR Manager both sign off on the extension request.
- Employees will be notified of extensions in writing at least 7 calendar days before the probation end date. 3. Impact on Benefits, Seniority, and Contract
- While on probation (including any extension), employees are not eligible for:
 - Annual leave encashment
 - Internal role transfers
 - Performance bonuses
- Seniority accrual starts only after successful probation completion.
- If an employee has two or more extensions in different roles (due to internal transfers), HR will assess contract renewal eligibility based on performance

Fig 28: Response generated when k=4 for ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Inference:

- In this case, ‘thenlper/gte-large’ has not addressed "Payroll and Termination" procedures, whereas the other two models stand out by retrieving appropriate chunks in response to the user query.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

For ‘thenlper/gte-large’ embedding model

Response:

```
Relevant Chunk 1:
5. Additional Conditions for Employees on Probation
• Special leave requests during probation are limited to:

  ◦ Maximum 2 consecutive working days for bereavement or emergency family care.

  ◦ Full jury duty leave is allowed if legally mandated, but probation period is automatically extended by the same duration.

• Approval requires both Supervisor and HR Manager sign-off.

6. Restrictions for Employees Under Harassment/Discrimination Investigation
• While under active investigation:

  ◦ Non-mandatory special leave (e.g., non-critical family care, discretionary events) is suspended until investigation concludes.

  ◦ Bereavement and jury duty exceptions still apply, but employee must provide live contact details during leave for investigation-related communication.
```

Fig 29: Response generated when k=4 for ‘thenlper/gte-large’ embedding model

For ‘BAAI/bge-large-en-v1.5’ embedding model

Response:

Relevant Chunk 1:

- Jury duty / legal summons: For the full duration specified in the official court document.
- Emergency family care: Up to 3 working days per calendar year.
- Natural disasters: Maximum 7 consecutive calendar days per incident.
- Additional days require conversion from annual leave or unpaid leave.

3. Documentation Requirements

Special leave is approved only upon submission of:

- Death certificate, funeral notice, or obituary (bereavement).
- Official court summons or jury duty letter (legal obligations).
- Medical certificate from a registered practitioner (emergency care).
- Government-issued disaster report or evacuation notice (natural disasters).
- All documents must be submitted within 5 working days of returning to duty.

Fig 30: Response generated when k=4 for ‘BAAI/bge-large-en-v1.5’ embedding model

For ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Response:

Relevant Chunk 1:

- Jury duty / legal summons: For the full duration specified in the official court document.
- Emergency family care: Up to 3 working days per calendar year.
- Natural disasters: Maximum 7 consecutive calendar days per incident.
- Additional days require conversion from annual leave or unpaid leave.

3. Documentation Requirements

Special leave is approved only upon submission of:

- Death certificate, funeral notice, or obituary (bereavement).
- Official court summons or jury duty letter (legal obligations).
- Medical certificate from a registered practitioner (emergency care).
- Government-issued disaster report or evacuation notice (natural disasters).
- All documents must be submitted within 5 working days of returning to duty.

Fig 31: Response generated when k=4 for ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Inference:

- Each of the three embedding models retrieved identical chunks that were relevant to the user query, albeit in a different order.

Query 3: What should I do if I notice suspected harassment with my female colleague?

For 'thenlper/gte-large' embedding model

Response:

```
Relevant Chunk 1:
• Accepted reporting channels:

1. Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).

2. Secure Email Portal: hr.ethics@flykiteair.com.

3. Anonymous drop-boxes in crew lounges and staff cafeterias.

• Reports must include date, location, persons involved, and a brief incident description.

4. Protection Against Retaliation
• Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.

• Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.

5. Investigation & Resolution Timelines
• HR must initiate a preliminary review within 3 working days of receiving a report
```

Fig 32: Response generated when k=4 for 'thenlper/gte-large' embedding model

For 'BAAI/bge-large-en-v1.5' embedding model

Response:

```

Relevant Chunk 1:
• Accepted reporting channels:

1. Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).

2. Secure Email Portal: hr.ethics@flykiteair.com.

3. Anonymous drop-boxes in crew lounges and staff cafeterias.

• Reports must include date, location, persons involved, and a brief incident description.

4. Protection Against Retaliation
• Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.

• Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.

5. Investigation & Resolution Timelines
• HR must initiate a preliminary review within 3 working days of receiving a report.

```

Fig 33: Response generated when k=4 for ‘BAAI/bge-large-en-v1.5’ embedding model

For ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Response:

```

Relevant Chunk 1:
• Accepted reporting channels:

1. Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).

2. Secure Email Portal: hr.ethics@flykiteair.com.

3. Anonymous drop-boxes in crew lounges and staff cafeterias.

• Reports must include date, location, persons involved, and a brief incident description.

4. Protection Against Retaliation
• Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.

• Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.

5. Investigation & Resolution Timelines
• HR must initiate a preliminary review within 3 working days of receiving a report

```

Fig 34: Response generated when k=4 for ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Inference:

- The 'nomic-ai/nomic-embed-text-V1.5' embedding model retrieved data unrelated to the user query, such as 'Data Protection and Privacy' and 'Attendance and Absence Management'. In contrast the other two embedding models retrieved relevant chunks related to the user query.

Fetching relevant chunks when k=8 for all three embedding models

Query 1: What are the effects on the benefits I receive if my probation is extended?

For 'thenlper/gte-large' embedding model

Response:

```
Relevant Chunk 1:
• Extensions are granted only if:
  a. The employee has achieved at least 60% of their probationary objectives.
  b. A written PIP with measurable targets is issued within 5 working days of the original probation end date.
  c. The Department Head and HR Manager both sign off on the extension request.

• Employees will be notified of extensions in writing at least 7 calendar days before the probation end date.
3. Impact on Benefits, Seniority, and Contract
• While on probation (including any extension), employees are not eligible for:
  o Annual leave encashment
  o Internal role transfers
  o Performance bonuses

• Seniority accrual starts only after successful probation completion.

• If an employee has two or more extensions in different roles (due to internal transfers), HR will assess contract renewal eligibility based on performance history. No automatic carry-over is granted.
```

Fig 35: Response generated when k=8 for 'thenlper/gte-large' embedding model

For 'BAAI/bge-large-en-v1.5' embedding model

Response:

Relevant Chunk 1:

- Extensions are granted only if:
 - a. The employee has achieved at least 60% of their probationary objectives.
 - b. A written PIP with measurable targets is issued within 5 working days of the original probation end date.
 - c. The Department Head and HR Manager both sign off on the extension request.
- Employees will be notified of extensions in writing at least 7 calendar days before the probation end date. 3. Impact on Benefits, Seniority, and Contract
- While on probation (including any extension), employees are not eligible for:
 - Annual leave encashment
 - Internal role transfers
 - Performance bonuses
- Seniority accrual starts only after successful probation completion.
- If an employee has two or more extensions in different roles (due to internal transfers), HR will assess contract renewal eligibility based on performance

Fig 36: Response generated when k=8 for ‘BAAI/bge-large-en-v1.5’ embedding model

For ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Response:

Relevant Chunk 1:

- Extensions are granted only if:
 - a. The employee has achieved at least 60% of their probationary objectives.
 - b. A written PIP with measurable targets is issued within 5 working days of the original probation end date.
 - c. The Department Head and HR Manager both sign off on the extension request.
- Employees will be notified of extensions in writing at least 7 calendar days before the probation end date. 3. Impact on Benefits, Seniority, and Contract
- While on probation (including any extension), employees are not eligible for:
 - Annual leave encashment
 - Internal role transfers
 - Performance bonuses
- Seniority accrual starts only after successful probation completion.
- If an employee has two or more extensions in different roles (due to internal transfers), HR will assess contract renewal eligibility based on performance

Fig 37: Response generated when k=8 for ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Inference:

- While all of the embedding models were returning the same chunks, some of them were unrelated to the user's inquiry.

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

For 'thenlper/gte-large' embedding model

Response:

```
Relevant Chunk 1:
5. Additional Conditions for Employees on Probation
• Special leave requests during probation are limited to:

  ◦ Maximum 2 consecutive working days for bereavement or emergency family care.

  ◦ Full jury duty leave is allowed if legally mandated, but probation period is automatically extended by the same duration.

• Approval requires both Supervisor and HR Manager sign-off.

6. Restrictions for Employees Under Harassment/Discrimination Investigation
• While under active investigation:

  ◦ Non-mandatory special leave (e.g., non-critical family care, discretionary events) is suspended until investigation concludes.

  ◦ Bereavement and jury duty exceptions still apply, but employee must provide live contact details during leave for investigation-related communication.
```

Fig 38: Response generated when k=8 for 'thenlper/gte-large' embedding model

For 'BAAI/bge-large-en-v1.5' embedding model

Response:

Relevant Chunk 1:

- Jury duty / legal summons: For the full duration specified in the official court document.
- Emergency family care: Up to 3 working days per calendar year.
- Natural disasters: Maximum 7 consecutive calendar days per incident.
- Additional days require conversion from annual leave or unpaid leave.

3. Documentation Requirements

Special leave is approved only upon submission of:

- Death certificate, funeral notice, or obituary (bereavement).
- Official court summons or jury duty letter (legal obligations).
- Medical certificate from a registered practitioner (emergency care).
- Government-issued disaster report or evacuation notice (natural disasters).
- All documents must be submitted within 5 working days of returning to duty.

Fig 39: Response generated when k=8 for ‘BAAI/bge-large-en-v1.5’ embedding model

For ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Response:

Relevant Chunk 1:

- Jury duty / legal summons: For the full duration specified in the official court document.
- Emergency family care: Up to 3 working days per calendar year.
- Natural disasters: Maximum 7 consecutive calendar days per incident.
- Additional days require conversion from annual leave or unpaid leave.

3. Documentation Requirements

Special leave is approved only upon submission of:

- Death certificate, funeral notice, or obituary (bereavement).
- Official court summons or jury duty letter (legal obligations).
- Medical certificate from a registered practitioner (emergency care).
- Government-issued disaster report or evacuation notice (natural disasters).
- All documents must be submitted within 5 working days of returning to duty.

Fig 40: Response generated when k=8 for ‘nomic-ai/nomic-embed-text-v1.5’ embedding model

Inference:

- Regardless of the embedding model employed, a combination of relevant and irrelevant chunks were retrieved based on the user query.

Query 3: What should I do if I notice suspected harassment with my female colleague?

For 'thenlper/gte-large' embedding model

Response:

```
Relevant Chunk 1:
• Accepted reporting channels:

1. Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).

2. Secure Email Portal: hr.ethics@flykiteair.com.

3. Anonymous drop-boxes in crew lounges and staff cafeterias.

• Reports must include date, location, persons involved, and a brief incident description.

4. Protection Against Retaliation
• Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.

• Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.

5. Investigation & Resolution Timelines
• HR must initiate a preliminary review within 3 working days of receiving a report
```

Fig 41: Response generated when k=8 for 'thenlper/gte-large' embedding model

For 'BAAI/bge-large-en-v1.5' embedding model

Response:

Relevant Chunk 1:

- Accepted reporting channels:

- Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).
- Secure Email Portal: hr.ethics@flykiteair.com.
- Anonymous drop-boxes in crew lounges and staff cafeterias.

- Reports must include date, location, persons involved, and a brief incident description.

- Protection Against Retaliation
 - Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.
 - Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.
- Investigation & Resolution Timelines
 - HR must initiate a preliminary review within 3 working days of receiving a report.

Fig 42: Response generated when k=8 for 'BAAI/bge-large-en-v1.5' embedding model

For 'nomic-ai/nomic-embed-text-v1.5' embedding model

Response:

Relevant Chunk 1:

- Accepted reporting channels:

- Confidential HR Helpline: +91-8000-456-789 (available Mon-Sat, 9 AM-8 PM IST).
- Secure Email Portal: hr.ethics@flykiteair.com.
- Anonymous drop-boxes in crew lounges and staff cafeterias.

- Reports must include date, location, persons involved, and a brief incident description.

- Protection Against Retaliation
 - Retaliation (e.g., demotion, shift change, exclusion from training) against a reporting employee is grounds for immediate disciplinary action.
 - Allegations of retaliation must be reported within 7 calendar days of occurrence, even if the original harassment/discrimination case is still under review.
- Investigation & Resolution Timelines
 - HR must initiate a preliminary review within 3 working days of receiving a report.

Fig 43: Response generated when k=8 for 'nomic-ai/nomic-embed-text-v1.5' embedding model

Inference:

- Regardless of the embedding model, every chunk that was retrieved was identical; nevertheless, some irrelevant chunks were also retrieved in accordance with the user query.

Final Inferences:

- Relevant document chunks are retrieved using the "Similarity" search strategy.
- By collecting relevant chunks in response to user queries, the 'BAAI/bge-large-en-v1.5' embedding model performs better.
- Very few contextual pieces were obtained when k=2 was lower. Some unnecessary chunks were obtained when the value of k=8 was higher. In order to ensure that appropriate chunks were obtained in accordance with user queries, any ideal value of k=4 is preferable.

Insights and Recommendations

Insights

- **Question Answering with LLM:** Among three user inquiries, one of the responses generated by the LLM hallucinated between two areas—criminal and corporate—due to the specific term "Probation" used in the user query. Additionally, the LLM provided generic responses instead of functioning as a specialized company bot.
- **Question Answering with LLM and Prompt Engineering:** Instances of cross-domain hallucinations have been successfully removed by the system prompt alongside the user query, allowing it to operate more like a corporate HR assistant, while still delivering basic answers without referencing corporate documentation.
- **Evaluation**
 - **Groundedness:** This is the extent to which the generated response is grounded to the context. However, the lack of context resulted in a very low score of 1 for the responses produced by both the LLM and LLM with prompt engineering.
 - **Relevance:** This is a measure of how closely the generated response relates to the user's query. In this case, the majority of the answers produced by the LLM and LLM with prompt engineering were rated between 4 and 5, meaning that they generally successfully addressed the user's query.
- **Data Preparation for RAG**
 - **Chunking:** A variety of chunking techniques were employed, including character, recursive character and semantic text splitters. The semantic method performed better because the document is chunked based on similarity between adjacent sentences, whereas the character text splitter splits words abruptly and without meaning, and the recursive character text splitter splits words fixedly but not abruptly.
 - **Embedding:** The sentences were transformed into dimensional vectors using three distinct embedding models: "thenlper/gte-large," "BAAI/bge-large-en-v1.5," and "nomic-ai/nomic-embed-text-V1.5."
 - **Vector Store:** To conduct quick similarity searches, document chunks are stored in embeddings in the "Chroma" database.

- **Retriever:** The 'similarity' approach is used to extract relevant document chunk embeddings stored in the database by converting user queries to embeddings.

Recommendations

- **Integrate Contextual Retrieval for LLM Responses:** By providing the LLM with relevant context, the poor groundedness scores and generic responses can be addressed.
- **Improve Prompt Engineering with Context Retrieved:** Although domain adherence was improved via prompt engineering, it can be further improved by directly incorporating the retrieved context into the prompt. The LLM should always base its responses solely on the context given by the system prompt.
- **Improved Embedding Model:** The 'BAAI/bge-large-en-v1.5' embedding model outperformed the others, followed by 'thenlper/gte-large' and 'nomic-ai/nomic-embed-text-V1.5', when all the embedding models were compared by looking at the recovered chunks as the user query. Chunks are therefore transformed into dimensional vectors using the 'BAAI/bge-large-en-v1.5' embedding model.
- **Optimal k value:** Since the content is chunked using the semantic technique, a value of $k=4$ is advised in order to ensure that only relevant chunks were fetched in response to user queries.

Appendix

Installing and Importing Necessary Libraries and Dependencies

```
# Installation for GPU llama-cpp-python
!CMAKE_ARGS="-DLLAMA_CUBLAS=on" FORCE_CMAKE=1 pip install llama-cpp-
python==0.1.85 --force-reinstall --no-cache-dir -q
# For installing the libraries & downloading models from HF hub
!pip install huggingface_hub==0.35.0 pandas==2.2.2 tiktoken==0.11.0 pymupdf==1.26.4
langchain==0.3.27 langchain-community==0.3.29 langchain_experimental==0.3.4 chromadb==1.1.0
sentence-transformers==5.1.0 numpy==2.3.3 -q
# Libraries for processing dataframes, text
import json, os
import tiktoken
import pandas as pd

# Libraries for loading data, chunking, Embedding and Vector Databases
from langchain.text_splitter import RecursiveCharacterTextSplitter, CharacterTextSplitter
from langchain_experimental.text_splitter import SemanticChunker
from langchain_community.document_loaders import PyMuPDFLoader
from langchain_community.embeddings.sentence_transformer import SentenceTransformerEmbeddings
from langchain_community.vectorstores import Chroma
```

```
# Library for downloading and loading llm
from huggingface_hub import hf_hub_download
from llama_cpp import Llama
```

Question Answering using LLM

Downloading and Loading the model

```
# Loading model
model_name_or_path = 'TheBloke/Mistral-7B-Instruct-v0.2-GGUF'
model_basename = 'mistral-7b-instruct-v0.2.Q6_K.gguf' # model in gguf format
# Using hf_hub_download to download a model from the Hugging face model hub
model_path = hf_hub_download(
    repo_id = model_name_or_path, # repo id specifies the model name or path in Hugging face
    repository
    filename = model_basename # specifies name of the file to download
)
# Initialize Llama model
llm = Llama(
    model_path = model_path,
    n_ctx = 4096, # context window
    n_batch = 512, # Should be between 1 and n_ctx
    n_gpu_layers = 38 # GPU VRAM pool
)
```

Function to define model parameters for generating response

```
# Function to generate response
def response(query, max_tokens=1024, temperature=0, repeat_penalty=1.2, top_p=0.95, top_k=50):
    model_output = llm(
        prompt=query, # User query
```



```

        max_tokens = max_tokens,          # Number of tokens the response should be generated by the
model
        temperature = temperature,        # Controls the randomness of the response to be generated
        top_p = top_p,                    # Controls the diversity of the generated response by establishing
cumulative probability cut-off for token selection
        top_k = top_k,                    # Maximum number of most likely next tokens to be considered
        repeat_penalty = repeat_penalty,  # Penalizes repeated tokens to improve text diversity
        stop = ['INST'],
        echo = False
    )

    return model_output['choices'][0]['text']

```

List of Questions

List of questions

```

user_queries = [
    'What are the effects on the benefits I receive if my probation is extended?',
    'There has been a demise in my family last night, and I need to attend the last rites. How should I
inform the office, and will I be granted leave?',
    'What should I do if I notice suspected harassment with my female colleague?'
]

```

Query 1: What are the effects on the benefits I receive if my probation is extended?

```

# get the response for the user query
query_1 = user_queries[0]
response_1 = response(query=query_1)
print(response_1)

```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```

# get the response for the user query
query_2 = user_queries[1]
response_2 = response(query=query_2)
print(response_2)

```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```

# get the response for the user query
query_3 = user_queries[2]
response_3 = response(query=query_3)
print(response_3)

```

Evaluating the response generated

System message for evaluating groundedness

```
groundedness_rater_system_message = ""
```

You are tasked to rate the AI generated answers to the questions posed by the employees.

You will be presented with the question, context used by the AI system to generate the answer and an AI generated answer to the question.

In the input the question will begin with ###Question, the context will begin with ###Context while the AI generated answer will begin with ###Answer.

Evaluation criteria:

Your task is to judge the extent to which the metric is followed by the answer.

Rate it 1 - The metric not followed at all

Rate it 2 - The metric is followed only to a limited extent

Rate it 3 - The metric is followed to a good extent

Rate it 4 - The metric is followed mostly

Rate it 5 - The metric is followed completely

Metric:

The answer should be strictly derived only from the information presented in the context.

Instructions:

1. First write down the steps that are needed to evaluate the answer as per the metric.
2. Give a step-by-step explanation if the answer adheres to the metric considering the question and context as the input.
3. Next, evaluate the extent to which the metric is followed.
4. Use the previous information to rate the answer using the evaluation metric and assign a score.

MANDATORY CRITICAL RULE:

If the text following the "###Context" tag is completely blank, empty, or missing, then NO information was provided to back up the answer. In this exact case, you MUST bypass all other instructions and automatically assign a Groundedness rating of 1. Do not give a 4 or a 5 if the context is empty.

Please note: Make sure you give a single overall rating in the range of 1 to 5 along with a brief explanation of why you assigned that score.

""""

System message for evaluating relevance

relevance_rater_system_message = """"

You are tasked to rate the AI generated answers to the questions posed by the employees.

You will be presented with the question, and an AI generated answer to the question.

In the input the question will begin with ###Question, while the AI generated answer will begin with ###Answer.

Evaluation criteria:

Your task is it judge the extent to which the metric is followed by the answer.

Rate it 1 - The metric not followed at all

Rate it 2 - The metric is followed only to a limited extent

Rate it 3 - The metric is followed to a good extent

Rate it 4 - The metric is followed mostly

Rate it 5 - The metric is followed completely

Metric:

Relevance measures how well the answer addresses the main aspects of the question.

Consider whether all and only important aspects are contained in the answer when evaluating relevance.

Instructions:

1. First write down the steps that are needed to evaluate the answer as per the metric.
2. Give step-by-step explanation if the answer adheres to the metric considering the question as the input.
3. Next, evaluate the extent to which the metric is followed.
4. Use the previous information to rate the answer using the evaluation metric and assign a score.

Please note: Make sure you give single overall rating in the range of 1 to 5 along with the brief explanation of why you assigned that score.

```
"""
# user template
user_message_template = """
###Question
{question}

###Context
{context}

###Answer
{answer}
"""
# Function for evaluating the responses generated by AI
def eval_grounded_relevance(question, answer, context, max_tokens=1024, temperature=0, top_p=0.95,
top_k=50, repeat_penalty=1.2):
    """
    A function to evaluate the groundedness and relevance of the response for the user query
    Key inputs required
    user_query: Question posed by the user
    response: Response generated by the llm
    context: Context used by the llm to generate the response
    """

    # Groundedness evaluation
    # If no context or empty context is provided else device groundedness prompt
    if not context or str(context).strip() == "":
        # Output of groundedness, if context is not presented
        final_grounded_response = "Score: 1 - Evaluation bypassed because ###Context is completely empty.
No data was provided to ground the answer."
    else:
        # Combine the user and system message to create a groundedness prompt
        groundedness_prompt = f"""[INST]{groundedness_rater_system_message}
        {'user': {user_message_template.format(question=question, context=context,
answer=answer)}}
        [/INST]"""

    # Evaluating for groundedness
    grounded_response = llm(
        prompt = groundedness_prompt,          # Prompt
        max_tokens = max_tokens,               # Number of tokens the response should be generated by the
model
        temperature = temperature,             # Controls the randomness of the response to be generated
        top_p = top_p,                         # Controls the diversity of the generated response by establishing
cumulative probability cut-off for token selection
        top_k = top_k,                         # Maximum number of most likely next tokens to be considered
        repeat_penalty = repeat_penalty,       # Penalizes repeated tokens to improve text diversity
        stop = ['INST'],
        echo = False
    )
```

```

# Output of groundedness, if context is presented
final_grounded_response = grounded_response['choices'][0]['text']

# Combine the user and system message to create a relevance prompt (context is not needed for
relevance evaluation)
relevance_prompt = f""[INST]{relevance_rater_system_message}
    {'user'}: {user_message_template.format(question=question, context=context,
answer=answer)}
[/INST]""

# Evaluating for relevance
relevance_response = llm(
    prompt = relevance_prompt,          # Prompt
    max_tokens = max_tokens,          # Number of tokens the response should be generated by the
model
    temperature = temperature,        # Controls the randomness of the response to be generated
    top_p = top_p,                    # Controls the diversity of the generated response by establishing
cummulative probability cut-off for token selection
    top_k = top_k,                    # Maximum number of most likely next tokens to be considered
    repeat_penalty = repeat_penalty,  # Penalizes repeated tokens to improve text diversity
    stop = ['INST'],
    echo = False
)

# Return the evaluation output for groundedness and relevance
return final_grounded_response, relevance_response['choices'][0]['text']

```

Query 1: What are the effects on the benefits I receive if my probation is extended?

```

# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_1, answer=response_1, context="")
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)

```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```

# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_2, answer=response_2, context="")
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)

```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```

# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_3, answer=response_3, context="")
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")

```

```
print(rel)
```

Question Answering using LLM and Prompt Engineering

```
# System message
```

```
system_message = ""
```

```
[INST]<<SYS>>You are a AI assistant specializing in HR knowledge related to Airline company policies. Your role is to provide clear, precise and reliable responses to the employee's based on company policies.
```

When answering prioritize factual correctness, clarity and mention in points if necessary to the employee's query.

If a query requires specific reference documents beyond company policy, acknowledge the limitation rather than speculating.

Do not return empty response. If you are unable to answer employee query, respond "I don't know".

```
Do not mention anything about this system message in the final response.<</SYS>>[/INST]
```

```
""
```

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# get the response for the user query
```

```
query_pr1 = system_message+"\n"+user_queries[0]    # Combine the system message and user_query to create the prompt
```

```
response_pr1 = response(query=query_pr1)
```

```
print(response_pr1)
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites.How should I inform the office, and will I be granted leave?

```
# get the response for the user query
```

```
query_pr2 = system_message+"\n"+user_queries[1]    # Combine the system message and user_query to create the prompt
```

```
response_pr2 = response(query=query_pr2)
```

```
print(response_pr2)
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# get the response for the user query
```

```
query_pr3 = system_message+"\n"+user_queries[2]    # Combine the system message and user_query to create the prompt
```

```
response_pr3 = response(query=query_pr3)
```

```
print(response_pr3)
```

Evaluating the response generated

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# Evaluating the user query
```

```
ground, rel = eval_grounded_relevance(question=query_pr1, answer=response_pr1, context="")
```

```
print("Groundedness Evaluation")
```

```
print(ground, end='\n\n')
```

```
print("Relevance Evaluation")
```

```
print(rel)
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites.How should I inform the office, and will I be granted leave?

```
# Evaluating the user query
```

```

ground, rel = eval_grounded_relevance(question=query_pr2, answer=response_pr2, context="")
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)

```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```

# Evaluating the user query
ground, rel = eval_grounded_relevance(question=query_pr3, answer=response_pr3, context="")
print("Groundedness Evaluation")
print(ground, end='\n\n')
print("Relevance Evaluation")
print(rel)

```

Data Preparation for RAG

Loading the Data

```

# Mount drive
from google.colab import drive
drive.mount('/content/drive')
# Get the PDF path
path = '/content/drive/MyDrive/AI & ML Course/12. Capstone Project/Dataset - Flykite Airlines_
HRP.pdf'

```

```

# Load the PDF data
pdf_data = PyMuPDFLoader(path)
air_hr = pdf_data.load()

```

Data Overview

Checking first 5 pages

```

# Looping through docs to check first 5 pages
for i in range(5):
    print(f'Page Number: {i+1}', end='\n')
    print(air_hr[i].page_content, end='\n\n')

```

Checking total pages

```

# Number of pages
len(air_hr)

```

Data Chunking

Character Text Splitter

```

# Chunking the document
char_text_splitter = CharacterTextSplitter(
    separator = " ", # Document separator
    chunk_size = 1000, # Maximum chunk size
    chunk_overlap = 20, # Number of tokens that overlap between the adjacent chunks
    strip_whitespace = False # Remove extra whitespaces
)

```

```

# Applying the text_splitter method on the pdf file
char_doc_chunks = pdf_data.load_and_split(char_text_splitter)

```

```
# Number of chunks the document is divided
len(char_doc_chunks)
```

```
# Checking a few document chunks from character chunking
print('Chunk 0:\n', char_doc_chunks[0].page_content, end='\n\n')
print('Chunk 1:\n', char_doc_chunks[1].page_content, end='\n\n')
print('Chunk 13:\n', char_doc_chunks[13].page_content, end='\n\n')
print('Chunk 14:\n', char_doc_chunks[14].page_content, end='\n\n')
print('Chunk 24:\n', char_doc_chunks[24].page_content, end='\n\n')
print('Chunk 25:\n', char_doc_chunks[25].page_content, end='\n\n')
```

Recursive Character Chunking based on Token Counts

```
# Chunking the document
rec_char_text_splitter = RecursiveCharacterTextSplitter.from_tiktoken_encoder(
    encoding_name = 'cl100k_base',
    chunk_size = 100,                # Maximum chunk size
    chunk_overlap = 20              # Number of tokens that overlap between the adjacent chunks
)
```

```
# Applying the text_splitter method on the pdf file
rec_char_doc_chunks = pdf_data.load_and_split(rec_char_text_splitter)
```

```
# Number of chunks the document is divided
len(rec_char_doc_chunks)
```

```
# Checking a few document chunks from recursive character chunking
print('Chunk 0:\n', rec_char_doc_chunks[0].page_content, end='\n\n')
print('Chunk 1:\n', rec_char_doc_chunks[1].page_content, end='\n\n')
print('Chunk 13:\n', rec_char_doc_chunks[13].page_content, end='\n\n')
print('Chunk 14:\n', rec_char_doc_chunks[14].page_content, end='\n\n')
print('Chunk 24:\n', rec_char_doc_chunks[24].page_content, end='\n\n')
print('Chunk 25:\n', rec_char_doc_chunks[25].page_content, end='\n\n')
```

Semantic Chunking

```
# Initialize embedding model(used to evaluate distance between sentences)
embeddings = SentenceTransformerEmbeddings(model_name='sentence-transformers/all-MiniLM-L6-
v2')
```

```
# Initialize semantic chunker
semantic_text_splitter = SemanticChunker(
    embeddings,                # Embedding model
    breakpoint_threshold_type='percentile',    # percentile method to break into chunks
    breakpoint_threshold_amount=95.0          # Splits at the 98th percentile of distance changes
)
```

```
# Creating chunks
sem_doc_chunks = pdf_data.load_and_split(semantic_text_splitter)
```

```
# Number of chunks the document is divided
len(sem_doc_chunks)
```

```
# Checking a few document chunks from semantic chunking
print('Chunk 0:\n', sem_doc_chunks[0].page_content, end='\n\n')
print('Chunk 1:\n', sem_doc_chunks[1].page_content, end='\n\n')
print('Chunk 13:\n', sem_doc_chunks[13].page_content, end='\n\n')
print('Chunk 14:\n', sem_doc_chunks[14].page_content, end='\n\n')
print('Chunk 24:\n', sem_doc_chunks[24].page_content, end='\n\n')
print('Chunk 25:\n', sem_doc_chunks[25].page_content, end='\n\n')
```

Embedding

Model: 'thenlper/gte-large'

```
# Loading the embedding model
```

```
gte_embedding_model = SentenceTransformerEmbeddings(model_name='thenlper/gte-large')
```

```
# Checking for embeddings for two different document chunks
```

```
gte_embedding_1 = gte_embedding_model.embed_query(sem_doc_chunks[20].page_content)
```

```
gte_embedding_2 = gte_embedding_model.embed_query(sem_doc_chunks[21].page_content)
```

```
print(f'Dimension of the embedding model: {len(gte_embedding_1)}')
```

```
len(gte_embedding_1) == len(gte_embedding_2)
```

Model: 'BAAI/bge-large-en-v1.5'

```
# Loading the embedding model
```

```
bge_embedding_model = SentenceTransformerEmbeddings(model_name='BAAI/bge-large-en-v1.5')
```

```
# Checking for embeddings for two different document chunks
```

```
bge_embedding_1 = bge_embedding_model.embed_query(sem_doc_chunks[20].page_content)
```

```
bge_embedding_2 = bge_embedding_model.embed_query(sem_doc_chunks[21].page_content)
```

```
print(f'Dimension of the embedding model: {len(bge_embedding_1)}')
```

```
len(bge_embedding_1) == len(bge_embedding_2)
```

Model: 'nomic-ai/nomic-embed-text-V1.5'

```
# Loading the embedding model
```

```
nom_embedding_model = SentenceTransformerEmbeddings(
```

```
    model_name='nomic-ai/nomic-embed-text-v1.5',
```

```
    model_kwargs={'trust_remote_code': True}
```

```
)
```

```
# Checking for embeddings for two different document chunks
```

```
nom_embedding_1 = nom_embedding_model.embed_query(sem_doc_chunks[20].page_content)
```

```
nom_embedding_2 = nom_embedding_model.embed_query(sem_doc_chunks[21].page_content)
```

```
print(f'Dimension of the embedding model: {len(nom_embedding_1)}')
```

```
len(nom_embedding_1) == len(nom_embedding_2)
```

Vector Database

```
# Create output directory
```

```
gte_out_dir = 'air_hr_man_gte'
```

```
bge_out_dir = 'air_hr_man_bge'
```

```
nom_out_dir = 'air_hr_man_nom'
```

```
# Function to embed the chunks and store them in a database
```

```
def create_store_embed(out_dir, embedding_model):
```

```
    """
```


Creates embeddings for document chunks and saves them into a specific directory.

out_dir The specific folder path for this database model.

embedding_model: Different embedding models.

"""

Create the specific model output directory if it doesn't exist

if not os.path.exists(out_dir):

os.makedirs(out_dir)

Database to store document chunks

vectorstore = Chroma.from_documents(

sem_doc_chunks, # Document chunks to be embedded

embedding_model, # Embedding function used to vectorize text chunks

persist_directory=out_dir # Saves the database to the folder

)

Load the existing stored data with the embedding model

vectorstore = Chroma(

persist_directory=out_dir, # Saved chunks

embedding_function=embedding_model # Embedding model used for user query

)

print(vectorstore.embeddings)

return vectorstore

create embeddings using 'thenlper/gte-large' model and store in database

gte_vectorstore = create_store_embed(out_dir=gte_out_dir, embedding_model=gte_embedding_model)

create embeddings using 'BAAI/bge-large-en-v1.5' model and store in database

bge_vectorstore = create_store_embed(out_dir=bge_out_dir, embedding_model=bge_embedding_model)

create embeddings using 'nomic-ai/nomic-embed-text-V1.5' model and store in database

nom_vectorstore = create_store_embed(out_dir=nom_out_dir,

embedding_model=nom_embedding_model)

Retriever

Retrieving the relevant document chunks

def retriever(vectorstore, k):

"""

A function to retrieve relevant document chunks using similarity search method

where number of relevant document chunks were varied

vectorstore: Vector database

k: Number of relevant document chunks to be retrieved

"""

retriever = vectorstore.as_retriever(

search_type = 'similarity', # Getting relevant documents using similarity search method

search_kwargs = {'k':k} # Getting top k relevant document chunks

)

return retriever

Fetching relevant chunks when k=2 for all three embedding models

Retrieving the relevant document chunks when 'thenlper/gte-large' embedding model is used

gte_retriever_2 = retriever(gte_vectorstore, 2)

```
# Retrieving the relevant document chunks when 'BAAI/bge-large-en-v1.5' embedding model is used
bge_retriever_2 = retriever(bge_vectorstore, 2)
```

```
# Retrieving the relevant document chunks when 'nomic-ai/nomic-embed-text-V1.5' embedding model is used
nom_retriever_2 = retriever(nom_vectorstore, 2)
```

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# Retrieving relevant chunks when 'thenlper/gte-large' is used
relevant_chunk = gte_retriever_2.get_relevant_documents(user_queries[0])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")
```

```
# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
relevant_chunk = bge_retriever_2.get_relevant_documents(user_queries[0])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")
```

```
# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
relevant_chunk = nom_retriever_2.get_relevant_documents(user_queries[0])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```
# Retrieving relevant chunks when 'thenlper/gte-large' is used
relevant_chunk = gte_retriever_2.get_relevant_documents(user_queries[1])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")
```

```
# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
relevant_chunk = bge_retriever_2.get_relevant_documents(user_queries[1])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")
```

```
# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
relevant_chunk = nom_retriever_2.get_relevant_documents(user_queries[1])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Retrieving relevant chunks when 'thenlper/gte-large' is used
relevant_chunk = gte_retriever_2.get_relevant_documents(user_queries[2])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")
```

```
# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
relevant_chunk = bge_retriever_2.get_relevant_documents(user_queries[2])
```

```

for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')

# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
relevant_chunk = nom_retriever_2.get_relevant_documents(user_queries[2])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')

```

Fetching relevant chunks when k=4 for all three embedding models

```

# Retrieving the relevant document chunks when 'thenlper/gte-large' embedding model is used
gte_retriever_4 = retriever(gte_vectorstore, 4)

```

```

# Retrieving the relevant document chunks when 'BAAI/bge-large-en-v1.5' embedding model is used
bge_retriever_4 = retriever(bge_vectorstore, 4)

```

```

# Retrieving the relevant document chunks when 'nomic-ai/nomic-embed-text-V1.5' embedding model is used
nom_retriever_4 = retriever(nom_vectorstore, 4)

```

Query 1: What are the effects on the benefits I receive if my probation is extended?

```

# Retrieving relevant chunks when 'thenlper/gte-large' is used
relevant_chunk = gte_retriever_4.get_relevant_documents(user_queries[0])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')

# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
relevant_chunk = bge_retriever_4.get_relevant_documents(user_queries[0])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')

# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
relevant_chunk = nom_retriever_4.get_relevant_documents(user_queries[0])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')

```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```

# Retrieving relevant chunks when 'thenlper/gte-large' is used
relevant_chunk = gte_retriever_4.get_relevant_documents(user_queries[1])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')

# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
relevant_chunk = bge_retriever_4.get_relevant_documents(user_queries[1])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')

# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
relevant_chunk = nom_retriever_4.get_relevant_documents(user_queries[1])

```

```
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')
```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```
# Retrieving relevant chunks when 'thenlper/gte-large' is used
relevant_chunk = gte_retriever_4.get_relevant_documents(user_queries[2])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')
```

```
# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
relevant_chunk = bge_retriever_4.get_relevant_documents(user_queries[2])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')
```

```
# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
relevant_chunk = nom_retriever_4.get_relevant_documents(user_queries[2])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')
```

Fetching relevant chunks when k=8 for all three embedding models

```
# Retrieving the relevant document chunks when 'thenlper/gte-large' embedding model is used
gte_retriever_8 = retriever(gte_vectorstore, 8)
```

```
# Retrieving the relevant document chunks when 'BAAI/bge-large-en-v1.5' embedding model is used
bge_retriever_8 = retriever(bge_vectorstore, 8)
```

```
# Retrieving the relevant document chunks when 'nomic-ai/nomic-embed-text-V1.5' embedding model is used
nom_retriever_8 = retriever(nom_vectorstore, 8)
```

Query 1: What are the effects on the benefits I receive if my probation is extended?

```
# Retrieving relevant chunks when 'thenlper/gte-large' is used
relevant_chunk = gte_retriever_8.get_relevant_documents(user_queries[0])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')
```

```
# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
relevant_chunk = bge_retriever_8.get_relevant_documents(user_queries[0])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')
```

```
# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
relevant_chunk = nom_retriever_8.get_relevant_documents(user_queries[0])
for i, doc in enumerate(relevant_chunk):
    print(f'Relevant Chunk {i+1}:\n{doc.page_content}\n')
```

Query 2: There has been a demise in my family last night, and I need to attend the last rites. How should I inform the office, and will I be granted leave?

```

# Retrieving relevant chunks when 'thenlper/gte-large' is used
relevant_chunk = gte_retriever_8.get_relevant_documents(user_queries[1])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")

# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
relevant_chunk = bge_retriever_8.get_relevant_documents(user_queries[1])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")

# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
relevant_chunk = nom_retriever_8.get_relevant_documents(user_queries[1])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")

```

Query 3: What should I do if I notice suspected harassment with my female colleague?

```

# Retrieving relevant chunks when 'thenlper/gte-large' is used
relevant_chunk = gte_retriever_8.get_relevant_documents(user_queries[2])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")

# Retrieving relevant chunks when 'BAAI/bge-large-en-v1.5' is used
relevant_chunk = bge_retriever_8.get_relevant_documents(user_queries[2])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")

# Retrieving relevant chunks when 'nomic-ai/nomic-embed-text-V1.5' is used
relevant_chunk = nom_retriever_8.get_relevant_documents(user_queries[2])
for i, doc in enumerate(relevant_chunk):
    print(f"Relevant Chunk {i+1}:\n{doc.page_content}\n")

```